Dyadic perturbations form an endlessly refinable parameter atlas

depth 1
$\Delta x = 1.750$



depth 2
$\Delta x = 0.875$



depth 3
$\Delta x = 0.438$



depth 4
$\Delta x = 0.219$



depth 5
$\Delta x = 0.109$



depth 1
$\Delta\theta = 22.500^\circ$



depth 2
$\Delta\theta = 11.250^\circ$



depth 3
$\Delta\theta = 5.625^\circ$



depth 4
$\Delta\theta = 2.812^\circ$



depth 5
$\Delta\theta = 1.406^\circ$



depth 1
$s = 1.09051$



depth 2
$s = 1.04427$



depth 3
$s = 1.02190$



depth 4
$s = 1.01089$



depth 5
$s = 1.00543$



The step size halves with depth. Finite compositions approximate bounded translations, rotations, and log-scales to arbitrary precision.